

GANGHUA WANG

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Address: 492 Ford Hall, 224 Church Street SE
Minneapolis, MN 55455

Research Interests: Privacy-preserving machine learning, deep learning theory and applications, machine learning theory, model fairness

EDUCATION

University of Minnesota, Twin Cities

Ph.D. in Statistics

Aug. 2019 - present

Minneapolis, MN

Advised by [Prof. Jie Ding](#) and co-advised by [Prof. Yuhong Yang](#)

Peking University

B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019

Beijing, China

EXPERIENCE

Cisco PhD internship [5]

Manager: Dr. Myungjin Lee, Cisco Systems Inc.

June 2022 - Aug. 2022

San Jose, CA

We revealed the relationship between local and global fairness metrics in the distributed learning, establishing the conditions under which locally fair training methods work well.

Cisco Research Fellow: Privacy-preserving machine learning [10, 6, 3]

Jan. 2021 - present

Advisor: Prof. Jie Ding and Prof. Yuhong Yang, University of Minnesota

Collaborated with the Cisco research team

Minneapolis, MN

We developed a foundational framework to understand and guide the design of model stealing attacks and defenses, with applications to machine-learning-as-a-service and model compression.

Research assistant: Deep learning theory [1, 7]

Advisor: Prof. Jie Ding, University of Minnesota

Sept. 2021 - Present

Minneapolis, MN

We proposed a new sparsity index to measure the compressibility of deep neural networks, which significantly improves the state-of-the-art deep pruning methods based on the Lottery Ticket Hypothesis.

Research assistant: Data privacy [4, 8]

Advisor: Prof. Jie Ding, University of Minnesota

June 2020 - Aug. 2020

Minneapolis, MN

We proposed a new framework to protect sensitive information in the data collection procedure, which faithfully retains the truth while still enabling efficient statistical inference.

PUBLICATIONS

Published

* indicates equal contributions

- [1] Enmao Diao*, **Ganghua Wang***, Jie Ding, Yuhong Yang, and Vahid Tarokh. “Pruning deep neural networks from a sparsity perspective”. In: *Proc. ICLR* (2023). [\[pdf\]](#).

- [2] Gen Li, **Ganghua Wang**, and Jie Ding. “Provable Identifiability of ReLU Neural Networks via LASSO Regularization”. In: *IEEE Trans. Inf. Theory* (2023).
- [3] Xun Xian*, **Ganghua Wang***, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Understanding backdoor attacks through the adaptability hypothesis”. In: *Proc. ICML* (2023). [\[pdf\]](#).
- [4] **Ganghua Wang**, Jie Ding, and Yuhong Yang. “Regression with Set-Valued Categorical Predictors”. In: *Statistica Sinica* (2022). [\[pdf\]](#).

Under review

- [5] **Ganghua Wang**, Ali Payani, Myungjin Lee, and Ramana Kompella. “Federated learning with group bias mitigation: beyond the local fairness”. In: *Proc. NeurIPS* (2023).
- [6] **Ganghua Wang***, Xun Xian*, Jayanth Srinivasa, Ashish Kundu, Xuan Bi, Mingyi Hong, and Jie Ding. “Demystifying Poisoning Backdoor Attacks from a Statistical Perspective”. In: *Proc. NeurIPS* (2023).
- [7] Wenjing Yang*, **Ganghua Wang***, Jie Ding, and Yuhong Yang. “A Theoretical Understanding of Neural Network Compression from Sparse Linear Approximation”. In: *arXiv preprint arXiv:2206.05604* (2022). [\[pdf\]](#).
- [8] **Ganghua Wang** and Jie Ding. “Subset Privacy: Draw from an Obfuscated Urn”. In: *arXiv preprint* (2021). [\[pdf\]](#).

Manuscript

- [9] Zhiyuan Tang, **Ganghua Wang**, and Jie Ding. “Classification with set-valued labels”. In: *Preparation* (2023).
- [10] **Ganghua Wang**, Jie Ding, and Yuhong Yang. “Model Privacy: A Framework to Understand Model Stealing Attack and Defense”. In: *Preparation* (2023).

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

School of Statistics Travel Award	2023
Summer Research Fellowship	2020
School of Statistics First Year Scholarship	2019

Awarded by Peking University

The Academic Excellence Scholarship (3 times)	2016 - 2018
Fang Zheng Scholarship	2017
Wu Si Scholarship	2016, 2018
Freshman Scholarship	2015

Awarded by Cisco Systems, Inc.

Cisco Research Graduate Award	2022
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TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

<i>STAT 4102 Theory of Statistics II</i>	Fall 2020
<i>STAT 3021 Introduction to Probability and Statistics</i>	Spring 2020
<i>STAT 3011 Introduction to Statistical Analysis</i>	Fall 2019

SERVICES

Reviewer of

Conference on Neural Information Processing Systems (NeurIPS)	2023
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The International Conference on Acoustics, Speech, & Signal Processing (ICASSP)	2022
Mentor of K-12 Outreach Program	
AEOP High School Apprenticeship K-12 Outreach Program to enhance diversity, equity, and inclusion (sponsored by Army Research Office)	2022

SOFTWARE

Python Packages

1. **SubsetPrivacy**: Implement the subset privacy mechanisms proposed in [8] and statistical inference methods for set-valued observations [4].
2. **ModelPrivacy**: Implement the common and proposed model stealing defense/attack strategies on benchmark datasets [10].
3. **FedGFT**: Implement a bias mitigation federated learning algorithm proposed in [5].

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB, R, SQL
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PRESENTATIONS AND TALKS

Joint Statistical Meeting	Aug. 2023
TBD	<i>Toronto, Canada</i>
International Conference on Econometrics and Statistics	Aug. 2023
Pruning deep neural networks from a sparsity perspective	<i>Tokyo, Japan</i>
International Conference on Machine Learning	July 2023
Understanding backdoor attacks through the adaptability hypothesis	<i>Honolulu, HI</i>
Center for Statistical Science, Peking University	June 2023
Invited talk	<i>Beijing, China</i>
International Conference on Learning Representations	May 2023
Pruning deep neural networks from a sparsity perspective	<i>Kigali, Rwanda</i>
School of Statistics Student Seminar	Mar. 2023
Machine learning safety	<i>Minneapolis, MN</i>